

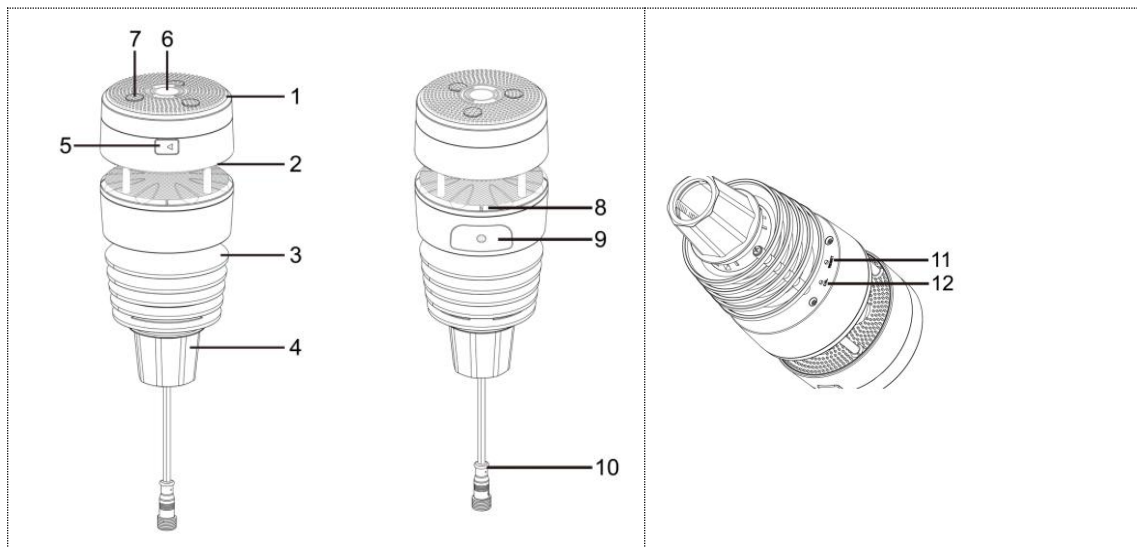
Weather Station with Modbus-RTU
Model: WN90LP

Manual

Features

- It will measure: temperature, humidity, rain volume, wind speed, wind direction, light, air pressure.
- Ultrasonic wind speed, wind direction
- Hapitac rain sensor
- With RS485 port (Modbus RTU)
- With cable: 4 wires cable, length: 10 meter
- Including 1 unit of Adaptor (RS485 to USB) (convenient to connect to computer)
- No display for it
- Communication protocol will be provided.

OVERVIEW:



1. solar panel (not included)	7. Hapitac Rainfall sensor
2. Ultrasonic wind sensor	8. NORTH alignment indicator
3. Temperature & humidity sensor	9. Battery compartment
4. Fixed Bolt (Mounting on a pole with 1 inch diameter)	10. cable connector
5. USB port (Factory use only)	11. Reset button
6. Light & UV sensor, LED indicator	12. Calibration button (factory use only)

Specification

* Sensor reporting interval: 8.8 seconds

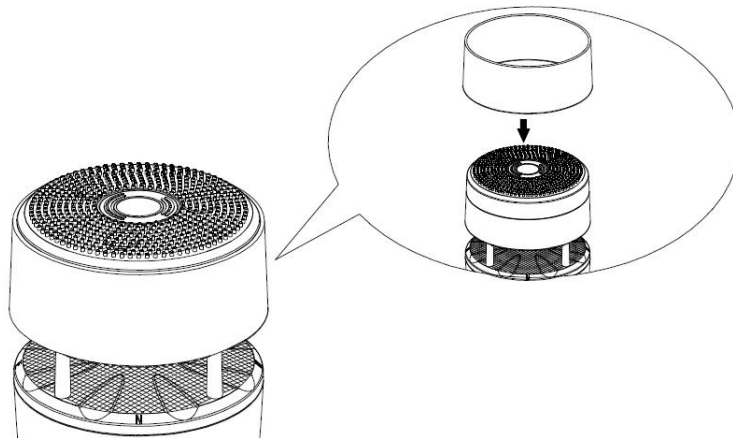
Measurement	Range	Accuracy	Resolution
Wind speed	0~40m/s	<10m/s, +/-0.5m/s ≥10m/s, +/-5%	0.1M/S
Wind direction	0~359°	<2m/s, ±10°, ≥2m/s, ±7°	1°
Rain	0 – 9999 mm	± 10%	0.1mm
Temperature	-40~60℃ (-40°F - 140°F)	±0.3℃ or ± 0.6°F	0.1℃ or 0.1°F
Humidity	1~99%	±3.5%	1%
Light	0~200Klux	±15%	10Lux
UVI	1~15	±2	1
Air pressure	300-1100hPa	+/-3hpa	1

Mount ultrasonic anemometer with piezoelectric assembly

Before you mount

Before installing your outdoor sensor in the permanent location, we recommend operating the device for one week in a temporary location with easy access. This will allow you to check out all of the functions, ensure proper operation and familiarize you with the device performance.

1. Put the white O-ring on the out of the upper part of outdoor sensor array for protection against moisture or water penetrated inside



2. Mounting on a pole as Figure. Make sure the mounting pole is vertical, or very close to it. Use a level as needed.

Note: Supports poles 1" OD

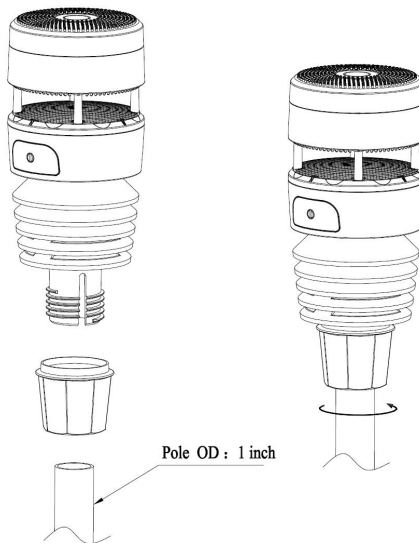


Figure : Sensor mounting diagram

3. There is an arrow icon with "N" words (as below Figure) representing the direction of North. The sensor body has to be adjusted so that the "N" indication is facing to real north direction in your location. A compass device is recommended to help adjust direction. Permanent wind direction error will be introduced when the outdoor sensor is not installed in right direction.

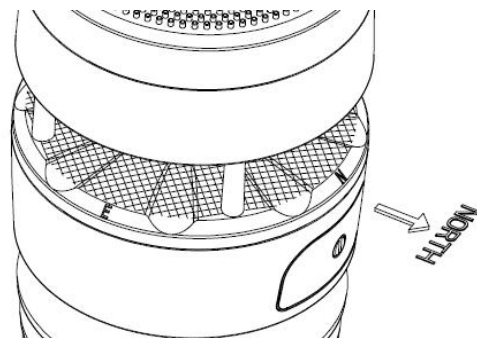


Figure: Facing North diagram

Note: In Southern hemisphere, it is not necessary to change the orientation to SOUTH.

Make sure the mounting tube for the sensor package is installed vertically (use a level at 90-degree offsets around the tube). Adjust the mounting pipe as necessary. Next also make sure the mounting of the anemometer body on the pipe is level. If it is not, wind direction and speed readings may not operate correctly or accurately. Adjust the mounting assembly as necessary.

4. During wintertime, to prevent ice formed on the reflection surface, the optional extension cord be connected to a 12v/1.0A outdoor type power adapter as Figure 6 showed below.

After power by the adaptor, the built-in thermostat heating plate will automatically turns on below 5°C (40°F) and aromatically turns off above 10°C (50°F), the heat plate will melts snow and ice and keeps its anemometer working as usual.

Note: The outdoor rated adaptor and extended power cable can be purchased separately:

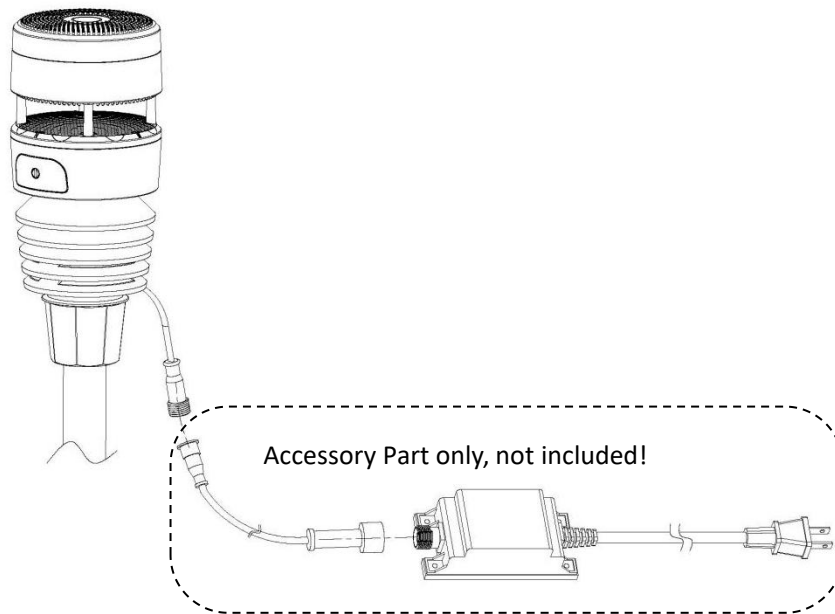


Figure 6: Sensor package mounting diagram

Reset Button and Transmitter LED

In the event the sensor package is not transmitting, reset the sensor.

Using a bent-open paperclip, press and hold the **RESET BUTTON** to affect a reset.

Connection (4 wires cable):

- Black wire: GND
- Red wire: USB A (D+)
- Yellow wire: USB B (D-)
- Green wire: VDD (4.5V -6V)



WN90LP Modbus RTU protocol V1.0.1

1. Communication Protocol

1.1 Basic Parameter

Code	8-bit binary
Data Bits	8 Bits
Parity Check Bits	None
Stop bit	1 bit
Error check	CRC(polynomial 0x8005)
Baud rate	Default:9600bit/s

1.2 Definition of data frame format

Host query frame structure:

Address code	Function code	Register start address	Register length	CRC LSB	CRC MSB
1 byte	1 byte	2 bytes	2 bytes	1 byte	1 byte

Slave reply frame structure:

Address code	function code	Number of valid bytes	Data set 1	Data set 2	Data set N	CRC LSB	CRC MSB
1 byte	1 byte	1 byte	2 bytes	2 bytes	2 bytes	1 byte	1 byte

1.3 Register address

Register address	Function	W/R	Description
0160 H	Device model	read only/RO	Device model code (90 H)
0161 H	Device Baud rate	Read write/RW	1 for 4800 2 for 9600 3 for 19200 4 for 115200
0162 H	Device address	Read write /RW	Range: 1~252
0163 H	Device ID MSB	read only/RO	

0164 H	Device ID LSB	read only/RO	
0165 H	Light	read only/RO	value in hex $\text{Light} = \text{value} * 10$ (Range: 0lux -> 300,000lux) If invalid fill with 0xFFFF
0166 H	UV index	read only/RO	value in hex UVI $\text{UVI} = \text{value} / 10$ (Range: 0 -> 150) If invalid fill with 0xFFFF
0167 H	temperature	read only/RO	value in hex $10.5\text{ C} = 1\text{F9h}$ $-10.5\text{ C} = 127\text{h}$ with 400 offset added $\text{value} = (\text{value} - 400) / 10$ (Range: -40.0C -> 60.0C) If invalid fill with 0x7ff
0168 H	Humidity	read only/RO	data in hex (Range: 1% - 99%) If invalid fill with 0xFFFF
0169 H	Wind speed	read only/RO	data in hex If invalid fill with 0xFFFF. $\text{Wind Speed} = \text{WIND value} * 0.1\text{m/s}$ (0~40m/s)
016A H	Gust Speed	read only/RO	data in hex If invalid fill with 0xFFFF. $\text{Gust Speed} = \text{GUST value} * 0.1\text{m/s}$ (0~40m/s)
016B H	Wind direction	read only/RO	value in hex (Range: 0° - 359°) If invalid fill with 0xFFFF
016C H	Rainfall	read only/RO	data in hex $\text{Rain} = \text{value} * 0.1\text{mm}$ $1.8\text{mm} = 12\text{H}$
016D H	ABS Pressure	read only/RO	Value in hex $\text{ABS} = \text{value} * 0.1\text{hPa}$ $1002.6\text{hPa} = 272\text{AH}$ If invalid fill with 0xFFFF

1.4 Example for communication protocol

1.4.1 Example for normal command

Example 1: read light intensity

Inquiry frame:

Address code	function code	Start address	Payload size	CRC LSB	CRC MSB
0x90	0x03	0x01 0x65	0x00 0x01	0x89	0x68

Reply frame:

Address code	Function code	Payload size	light intensity	CRC LSB	CRC MSB
0x90	0x03	0x02	0x07 0xB0	0x46	0x1D

It means: the intensity is: 1968 (LUX)

Example 2: get data (light, UVI, Temperature, Humidity, Wind Speed, Gust speed, Wind direction, Rainfall)

Inquiry frame:

Address code	Function code	Start address	Payload size	CRC LSB	CRC MSB
0x90	0x03	0x01 0x65	0x00 0x08	0x49	0x6E

Reply frame:

Address code	function code	Payload size	Payload	CRC LSB	CRC MSB
0x90	0x03	0x10	0x06E7, 0x000D 0x0296, 0x003C 0x0000, 0x0000 0x0096, 0x0000	0xBD	0x2F

It means:

Light = 17670 Lux

UV index (UVI) = 1.3

Temperature = 26.2°C

Humidity = 60%

Wind speed = 0 m/s

Gust speed = 0 m/s

Wind direction = 150°

Rainfall = 0 mm

Example 3: modify Baud rate as 4800

Inquiry frame:

Address code	Function code	Register address	data	CRC LSB	CRC MSB
0x90	0x06	0x01 0x61	0x00 0x01	0x04	0xA9

Reply frame:

Address code	Function code	Payload size	Payload	CRC LSB	CRC MSB
0x90	0x06	0x02	0x00 0x01	0x84	0x95

Example 4: modify device address as 0x34

Inquiry frame:

Address code	Function code	Register address	data	CRC LSB	CRC MSB
0x90	0x06	0x01 0x62	0x00 0x34	0x34	0xBE

Reply frame:

Address code	Function code	Payload size	Payload	CRC LSB	CRC MSB
0x90	0x06	0x02	0x00 0x34	0x44	0x82

1.4.2 Special Command

If users modified device address and Baud rate, but forget the device address (modified) and baud rate(modified), then use this special command.

The structure of the host inquiry frame as following:

prefix	Read/write Baud rate	Read/Write device address	CRC LSB	CRC MSB
3 bytes	1 byte	1 byte 0: read device	1 byte	1 byte

Fix value 0xFDFDFD	0: read Baud rate 1: set Baud rate4800 2: set Baud rate9600 3: set Baud rate19200 4: set Baud rate115200	address 1~252: set de vice address		
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Structure of Slave Reply frame:

prefix	Baud rate	Device address	CRC LSB	CRC MSB
3 bytes Fix value 0xFDFDFD	1 byte 1: for Baud rate4800 2: for Baud rate9600 3: for Baud rate19200 4: for Baud rate115200	1 byte	1 byte	1 byte

Example 5: read Baud rate and slave address:

Inquiry frame:

prefix	Read Baud rate	Read device address	CRC LSB	CRC MSB
0xFDFDFD	0x00	0x00	0xE9	0x88

Reply frame:

prefix	Baud rate	Device address	CRC LSB	CRC MSB
0xFDFDFD	0x01	0x90	0xE8	0x74

It means: Baud rate is 4800, Slave device address is 0x90。

Example 6: set Baud rate as 9600:

Inquiry Frame:

prefix	Read Baud rate	Read device address	CRC LSB	CRC MSB
0xFDFDFD	0x02	0x00	0xE8	0xE8

Reply frame:

prefix	Baud rate	Device address	CRC LSB	CRC MSB
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0xFDFDFD	0x02	0x90	0xE8	0x84
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It means: set Baud rate as 9600 successfully, and read the slave device address 0x90.

Example 7: set device address as 0x01:

Inquiry frame:

prefix	Read Baud rate	Read device address	CRC LSB	CRC MSB
0xFDFDFD	0x00	0x01	0x28	0x48

Reply frame:

prefix	Baud rate	Device address	CRC LSB	CRC MSB
0xFDFDFD	0x02	0x01	0x29	0x28

It means: set device address as 0x01 successfully, and read Baud rate 9600

1.5 Error Code

Error Codes	Name	Description
01	illegal function	function code not 0x03、0x06
02	illegal data address	Register address not in the range
03	illegal data value	Data length cross the border/value wrong
08	CRC Verification failed	CRC Verification failed

Error code of Reply frame need add code 0x80, as example 8:

Example 8: Reply frame

Address code	Function code	Error code	CRC LSB	CRC MSB
0x90	0x83	0x08	0x11	0x1B